Statistical Learning I

Tutorial: Session 3
OLS, Ridge, LASSO, Elastic Net

August 7, 2026

The organising question

Session 1 changed the loss. Session 3 keeps squared-error loss and adds a penalty.

One question runs through the whole session

Which procedures depend on the units and the coordinate system in which $\mathbf{X}$ happens to be recorded?

- **OLS** depends only on the column space of $\mathbf{X}$ $\Rightarrow$ invariant under $\mathbf{X} \mapsto \mathbf{XA}$, any non-singular $\mathbf{A}$.
- **Ridge / LASSO / elastic net** penalise the coordinates of $\boldsymbol{\beta}$ $\Rightarrow$ not invariant.

That single contrast is why standardisation is optional for OLS and compulsory afterwards.

Setup and notation

Model: $Y_i = \alpha + \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i,\ i = 1, \ldots, n$, with $\mathbf{X} \in \mathbb{R}^{n \times p}$.

Centred quantities

$$\bar{y} = \tfrac{1}{n} \textstyle\sum_i Y_i, \qquad \mathbf{y}_c = \mathbf{y} - \bar{y}\mathbf{1}$$
$$\mathbf{X}_c = \mathbf{X} - \mathbf{1}\bar{\mathbf{x}}^\top$$

Three transformations of X

- general: $\mathbf{XA}$, $\mathbf{A}$ non-singular
- diagonal: $\mathbf{XD}$, $\mathbf{D} = \operatorname{diag}(d_j)$ (= change of units)
- whitening: $\mathbf{A}$ with $(\mathbf{X}_c\mathbf{A})^\top(\mathbf{X}_c\mathbf{A}) = \mathbf{I}$

Convention

The intercept is never penalised. Centre both sides, fit without an intercept, restore $\hat{\alpha} = \bar{y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}$.

Why centring leaves the slopes unchanged

$D = [\mathbf{1}\,\mathbf{X}]$, $\theta = (\alpha, \boldsymbol{\beta})^\top$. Normal equations $D^\top D\,\theta = D^\top \mathbf{y}$ in block form:

$$\begin{pmatrix} n & n\bar{\mathbf{x}}^\top \\ n\bar{\mathbf{x}} & \mathbf{X}^\top\mathbf{X} \end{pmatrix} \begin{pmatrix} \alpha \\ \boldsymbol{\beta} \end{pmatrix} = \begin{pmatrix} n\bar{y} \\ \mathbf{X}^\top\mathbf{y} \end{pmatrix}$$

Row 1 $\Rightarrow$ $\alpha = \bar{y} - \bar{\mathbf{x}}^\top\boldsymbol{\beta}$. Substitute into row 2 to eliminate α:

$$n\bar{\mathbf{x}}(\bar{y} - \bar{\mathbf{x}}^\top\boldsymbol{\beta}) + \mathbf{X}^\top\mathbf{X}\,\boldsymbol{\beta} = \mathbf{X}^\top\mathbf{y} \implies (\mathbf{X}^\top\mathbf{X} - n\bar{\mathbf{x}}\bar{\mathbf{x}}^\top)\boldsymbol{\beta} = \mathbf{X}^\top\mathbf{y} - n\bar{\mathbf{x}}\bar{y}.$$

Since $\mathbf{X}_c^\top\mathbf{X}_c = \mathbf{X}^\top\mathbf{X} - n\bar{\mathbf{x}}\bar{\mathbf{x}}^\top$ and $\mathbf{X}_c^\top\mathbf{y}_c = \mathbf{X}^\top\mathbf{y} - n\bar{\mathbf{x}}\bar{y}$, this *is* $\mathbf{X}_c^\top\mathbf{X}_c\,\boldsymbol{\beta} = \mathbf{X}_c^\top\mathbf{y}_c$ — the no-intercept normal equations of $\mathbf{y}_c$ on $\mathbf{X}_c$.

Consequence

$\boldsymbol{\beta}$ solves the same equations with or without α in the model $\Rightarrow$ centring cannot change the slopes; α merely absorbs the means. (Frisch–Waugh–Lovell, partition $= \{\mathbf{1}, \mathbf{X}\}$.)

OLS: three facts

1. Normal equations

$$\hat{\beta}^{\mathrm{OLS}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}, \qquad \hat{\mathbf{y}} = \mathbf{H}\mathbf{y} \text{ with } \mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top.$$

2. Centring $\Rightarrow$ intercept separates (previous slide)

$$\hat{\beta}_{\text{centred, no intercept}} = \hat{\beta}_{\text{slopes}}, \qquad \hat{\alpha} = \bar{y} - \bar{\mathbf{x}}^\top\hat{\beta}, \qquad \hat{\mathbf{y}} = \bar{y}\mathbf{1} + \mathbf{X}_c\hat{\beta}.$$

3. Invariance under $\mathbf{X} \mapsto \mathbf{X}\mathbf{A}$

$$\mathbf{X}\mathbf{A}\big((\mathbf{X}\mathbf{A})^\top\mathbf{X}\mathbf{A}\big)^{-1}(\mathbf{X}\mathbf{A})^\top = \mathbf{X}\mathbf{A}\,\mathbf{A}^{-1}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{A}^{-\top}\mathbf{A}^\top\mathbf{X}^\top = \mathbf{H}.$$

$$\Rightarrow \quad \hat{\mathbf{y}} \text{ unchanged}, \qquad \hat{\beta}_{\mathbf{A}} = \mathbf{A}^{-1}\hat{\beta}.$$

Why $\mathbf{H}$ is the whole story

- $\mathbf{A}$ non-singular $\Rightarrow \text{col}(\mathbf{XA}) = \text{col}(\mathbf{X})$.
- Least squares = orthogonal projection onto $\text{col}(\mathbf{X})$.
- Same subspace $\Rightarrow$ same projection $\Rightarrow$ same $\hat{\mathbf{y}}$; only the coordinates of that projection change.

Consequences

- Rescaling x_j rescales $\hat{\beta}_j$ and nothing else.
- Nobody standardises before OLS — there is nothing to gain.
- R^2, residuals, fitted values, $\hat{\sigma}^2$: all invariant.

This argument breaks the moment a penalty is added. $\longrightarrow$ next.

Ridge regression

Definition (on centred data)

$$\hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda) \;=\; \arg\min_{\boldsymbol{\beta}} \;\|\mathbf{y}_c - \mathbf{X}_c\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2 \;=\; (\mathbf{X}_c^\top\mathbf{X}_c + \lambda\mathbf{I})^{-1}\mathbf{X}_c^\top\mathbf{y}_c.$$

- $\mathbf{X}_c^\top\mathbf{X}_c + \lambda\mathbf{I} \succ 0$ for every $\lambda > 0$, even when $\mathbf{X}_c^\top\mathbf{X}_c$ is singular $\Rightarrow$ unique solution when OLS has none ($p > n$, collinearity).
- λ is chosen by cross-validation (balancing predictive risk minimization against model parsimony).

Not invariant under $\mathbf{X} \mapsto \mathbf{X}\mathbf{A}$

$$\left(\mathbf{A}^\top\mathbf{X}_c^\top\mathbf{X}_c\mathbf{A} + \lambda\mathbf{I}\right)^{-1}\mathbf{A}^\top\mathbf{X}_c^\top\mathbf{y}_c$$

The penalty $\lambda\mathbf{I}$ is written in the new coordinates; undoing the change of basis would need $\lambda(\mathbf{A}^\top\mathbf{A})^{-1}$, not $\lambda\mathbf{I}$. The $\mathbf{A}$'s do not cancel.

Standardisation: what it does and does not fix

Standardising is itself a transformation $\mathbf{X} \mapsto \mathbf{XD}$ with $\mathbf{D} = \mathrm{diag}(1/s_j)$ — diagonal, hence special.

	diagonal A (units)	**general A** (rotation)
OLS	invariant	invariant
ridge on raw $\mathbf{X}$	not invariant	not invariant
ridge on standardised $\mathbf{X}$	invariant	not invariant

The honest statement

Standardising makes ridge immune to the arbitrary choice of units, not to a change of basis.

Why it matters practically: without it, a predictor measured in small units gets a large $\hat{\beta}_j$ and is therefore penalised harder, for no statistical reason.

Whitening: $\mathbf{Z}^\top\mathbf{Z} = \mathbf{I}$

Construction

SVD $\mathbf{X}_c = \mathbf{U}\mathbf{D}\mathbf{V}^\top$. Take $\mathbf{A} = \mathbf{V}\mathbf{D}^{-1}$. Then $\mathbf{Z} = \mathbf{X}_c\mathbf{A} = \mathbf{U}$ and $\mathbf{Z}^\top\mathbf{Z} = \mathbf{I}$.

Requires $\mathbf{X}_c$ of full column rank (all $d_j > 0$).

Ridge becomes transparent

$$\hat{\beta}^{\text{ridge}}(\lambda) = (\mathbf{Z}^\top\mathbf{Z} + \lambda\mathbf{I})^{-1}\mathbf{Z}^\top\mathbf{y}_c = \big((1+\lambda)\mathbf{I}\big)^{-1}\mathbf{Z}^\top\mathbf{y}_c = \frac{\hat{\beta}_{\mathbf{Z}}^{\text{OLS}}}{1+\lambda}.$$

$\hat{\beta}_{\mathbf{Z}}^{\text{OLS}} := \mathbf{Z}^\top\mathbf{y}_c$ is the OLS coefficient in the whitened coordinates $\mathbf{Z}$ (since $\mathbf{Z}^\top\mathbf{Z} = \mathbf{I}$) — not the original $\hat{\beta}^{\text{OLS}}$ on $\mathbf{X}_c$.

- Every coefficient shrunk by the same factor $1/(1+\lambda)$.
- $\lambda = 0 \Rightarrow$ OLS; $\lambda \to \infty \Rightarrow 0$.
- Never exactly zero for finite $\lambda \Rightarrow$ ridge shrinks but does not select.

Teaching device, not a recipe: available only when $\mathbf{X}_c^\top\mathbf{X}_c$ is invertible, i.e. when ridge is least needed.

Coordinate-wise standardisation: the version that always works

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}, \qquad s_j = \mathrm{sd}(x_{\cdot j}).$$

Whitening

needs $\mathbf{X}_c^\top \mathbf{X}_c$ invertible

fails if $p > n$, duplicated or collinear columns

Coordinate-wise

needs only $s_j > 0$

always available

Interpretation

After standardising, every $\hat{\beta}_j$ answers the same question: *how much does Y move when x_j moves by one of its own standard deviations?*

Coefficients become comparable across predictors: smaller coefficient = smaller effect. On the raw scale that is false — the size of $\hat{\beta}_j$ is an artefact of the units.

Back-transform: $\hat{\beta}_j^{\text{raw}} = \hat{\beta}_j^{\text{std}} / s_j$, same fitted values.

LASSO

Definition

$$\hat{\boldsymbol{\beta}}^{\text{lasso}}(\lambda) = \arg\min_{\boldsymbol{\beta}} \; \|\mathbf{y}_c - \mathbf{X}_c\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1, \qquad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

- No closed form in general (contrast ridge). Solved by coordinate descent.
- The ℓ_1 ball $\{\|\boldsymbol{\beta}\|_1 \le t\}$ has corners on the coordinate axes; the solution of the constrained problem frequently lands on one.
- A corner is exactly a point where some $\beta_j = 0 \;\Rightarrow\;$ fitting and selection in one step.
- Weaker than ridge's invariance: only signed permutations of the axes (relabel/sign-flip, not rotate) preserve the fit — the ℓ_1 ball's corners sit on the axes, and a rotation moves them off. The same orthogonal **A** that left ridge unchanged generally does change the LASSO fit $\Rightarrow$ standardise first.

Geometry, not thresholding

The zeros come from the shape of the constraint set, not from any rule applied afterwards.

Two formulations, one solution

Constrained (bound) form

$$\min_{\boldsymbol{\beta}} \ \mathrm{RSS}(\boldsymbol{\beta}) \quad \text{s.t.} \quad \|\boldsymbol{\beta}\|_1 \le t$$

Lagrangian (penalised) form

$$\min_{\boldsymbol{\beta}} \ \mathrm{RSS}(\boldsymbol{\beta}) + \lambda \|\boldsymbol{\beta}\|_1$$

Penalised $\Rightarrow$ constrained: two lines, no duality needed

Let $\hat{\boldsymbol{\beta}}$ minimise $\mathrm{RSS}(\boldsymbol{\beta}) + \lambda\|\boldsymbol{\beta}\|_1$, and define $t := \|\hat{\boldsymbol{\beta}}\|_1$. For any $\boldsymbol{\beta}$ with $\|\boldsymbol{\beta}\|_1 \le t$,

$$\mathrm{RSS}(\boldsymbol{\beta}) + \lambda\|\boldsymbol{\beta}\|_1 \ \ge \ \mathrm{RSS}(\hat{\boldsymbol{\beta}}) + \lambda\|\hat{\boldsymbol{\beta}}\|_1 \implies \mathrm{RSS}(\boldsymbol{\beta}) \ \ge \ \mathrm{RSS}(\hat{\boldsymbol{\beta}}) + \lambda \underbrace{\left(\|\hat{\boldsymbol{\beta}}\|_1 - \|\boldsymbol{\beta}\|_1\right)}_{\ge 0} \ \ge \ \mathrm{RSS}(\hat{\boldsymbol{\beta}}).$$

So $\hat{\boldsymbol{\beta}}$ already solves the constrained problem at that t.

Converse

For $0 \le t < \|\hat{\boldsymbol{\beta}}^{\mathrm{OLS}}\|_1$ the constraint is active and the ℓ_1 ball has non-empty interior (Slater's condition), so a multiplier $\lambda \ge 0$ exists giving the same solution. Hence the two forms trace the same path:

$$\lambda = 0 \iff t = \|\hat{\boldsymbol{\beta}}^{\mathrm{OLS}}\|_1, \qquad \lambda \to \infty \iff t \to 0, \qquad t(\lambda) = \|\hat{\boldsymbol{\beta}}(\lambda)\|_1 \text{ continuous, non-increasing.}$$

Which form to think in

Constrained form: for *seeing*

There is a visible budget. Shrinkage is literally the ball closing in, and the zeros are the corners of that ball. Every geometric picture in this session lives here.

Lagrangian form: for *computing*

This is what solvers minimise, what the KKT conditions describe, and what gives soft-thresholding. But there is no budget on screen — the penalty just tilts the surface.

The correspondence is data-dependent

$t(\lambda) = \|\hat{\boldsymbol{\beta}}(\lambda)\|_1$ depends on $\mathbf{X}_c$ and $\mathbf{y}_c$: the same λ gives a different t on different data. So λ is not transferable between data sets, and must be cross-validated on the data at hand. (The soft-thresholding formula on the next slide is stated for the Lagrangian form; in the constrained form the same operator appears, but the threshold is the multiplier $\lambda^*(t)$ solving $\|S(\mathbf{b}, \lambda^*)\|_1 = t$ — it cannot be read off t directly.)

Both pictures, on the same data at matched (λ, t): `Session3_Shrinkage_Visualiser.html`, tab 3.

Orthonormal design: ridge multiplies, LASSO subtracts

With $\mathbf{Z}^\top \mathbf{Z} = \mathbf{I}$ the problem separates across coordinates:

Ridge

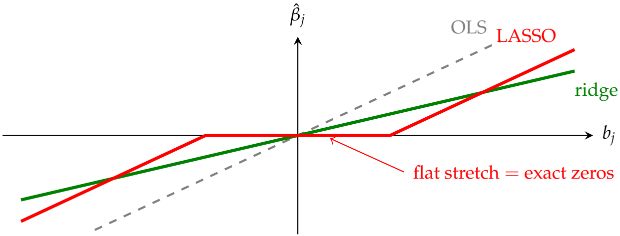
$$\hat{\beta}_j = \frac{b_j}{1+\lambda}$$

multiplication by a positive constant
$\Rightarrow$ can never cross 0

LASSO

$$\hat{\beta}_j = \operatorname{sign}(b_j)\big(|b_j| - t\big)_+$$

subtraction of a constant
$\Rightarrow$ crosses 0, and stays



b_j is the jth OLS coefficient; t is the penalty threshold parameter (scaling with $n\lambda$ under sample-averaged loss).

Coefficient paths

Ridge

- all p coefficients stay non-zero
- approach 0 asymptotically as $\lambda \to \infty$
- paths are smooth

LASSO

- coefficients hit 0 exactly, one by one
- path is piecewise linear in λ
- # non-zeros decreases with λ

Reading a path plot

x-axis displays $\log \lambda$ (or $\|\hat{\beta}\|_1$); each curve represents a predictor coefficient; vertical lines mark optimal penalty choices selected via cross-validation.

Caution. $\hat{\lambda}$ and the selected set are both random. The selected set is the less stable of the two — re-run CV with different folds before reporting "the" LASSO model.

Elastic net

Definition

$$\hat{\beta}^{\text{enet}} = \arg\min_{\beta} \frac{1}{2n}\|\mathbf{y}_c - \mathbf{X}_c\boldsymbol{\beta}\|_2^2 + \lambda\left[\alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2\right]$$

$$\alpha = 1 \;\Rightarrow\; \text{LASSO}, \qquad \alpha = 0 \;\Rightarrow\; \text{ridge}, \qquad 0 < \alpha < 1 \;\Rightarrow\; \text{mixture.}$$

Two LASSO weaknesses it repairs

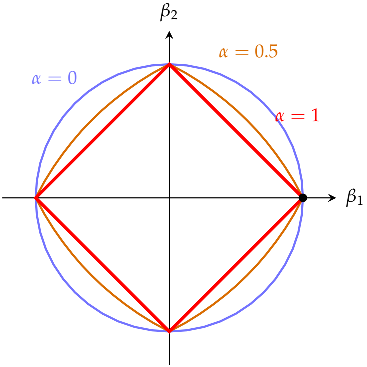
1. Among highly correlated predictors LASSO picks one nearly arbitrarily and zeroes the rest — unstable across samples.
2. When $p > n$, LASSO selects at most n predictors.

Formally (Zou & Hastie, 2005): for standardised predictors with correlation $\rho = \mathbf{x}_i^\top \mathbf{x}_j$ and $\hat{\beta}_i\hat{\beta}_j > 0$,

$$\frac{|\hat{\beta}_i - \hat{\beta}_j|}{\|\mathbf{y}\|_1} \le \frac{1}{\lambda_2}\sqrt{2(1-\rho)} \xrightarrow[\rho\to 1]{} 0.$$

The elastic net still selects: the corner survives

The penalty ball $\{\alpha\|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2 \le s\}$ is not differentiable on the axes for any $\alpha > 0$.



Why zeros survive: the KKT condition

$$\hat{\beta}_j = 0 \iff \left|\frac{1}{n}\mathbf{x}_j^\top(\mathbf{y}_c - \mathbf{X}_c\hat{\boldsymbol{\beta}})\right| \le \lambda\alpha$$

- The ℓ_2 term is differentiable at 0 with zero derivative, so it contributes nothing to this interval.
- Only the ℓ_1 term creates it, with half-width $\lambda\alpha$ — strictly positive for every $\alpha > 0$.

The corner, quantitatively

At the vertex the two one-sided tangents have slopes $\mp 1/\alpha$, so the interior angle is $\arccos\left(\frac{\alpha^2-1}{\alpha^2+1}\right)$:

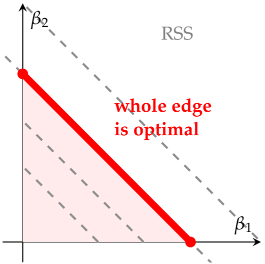
α	1	0.5	0.25	$\to 0$
angle	90°	127°	152°	180°

Selection weakens continuously as $\alpha \downarrow 0$, and vanishes only in the limit.

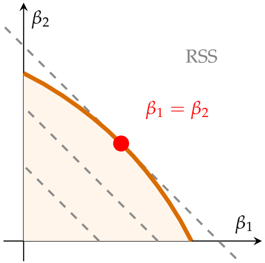
Grouping effect: contours parallel to the edge

Two identical standardised predictors, $\mathbf{x}_1 = \mathbf{x}_2$. Then $\mathbf{X}_c\boldsymbol{\beta} = \mathbf{x}_1(\beta_1 + \beta_2)$: the fit depends on $\boldsymbol{\beta}$ only through $\beta_1 + \beta_2$. So $\mathbf{X}_c^\top \mathbf{X}_c$ is singular and the RSS contours degenerate into parallel straight lines of slope -1.

LASSO: edge $\|$ contours



Elastic net: one tangency



LASSO: the tie cannot be broken

The active edge $\beta_1 + \beta_2 = t$ has slope -1 — exactly parallel to the contours. The whole edge attains the same RSS, so the solution set is a segment, and the solver returns an arbitrary point of it. A resample moves that point.

Elastic net: strict convexity picks the middle

The ℓ_2 term bows the edge outwards, so no arc of the boundary is parallel to the contours: the touching point is unique. By symmetry it is $\beta_1 = \beta_2 = c/2$ — among all splits with $\beta_1 + \beta_2 = c$, the quantity $\beta_1^2 + \beta_2^2$ is uniquely minimised there.

Summary

	OLS	Ridge	LASSO	Elastic net
penalty	—	$\|\|\boldsymbol{\beta}\|\|_2^2$	$\|\|\boldsymbol{\beta}\|\|_1$	both
closed form	yes	yes	no	no
defined if $p > n$	no	yes	yes	yes
$\hat{\boldsymbol{\beta}}$ unique	if $p < n$	always	†	always
shrinks	no	yes	yes	yes
exact zeros	no	no	yes	yes
invariant, **XA**	yes	no	no	no
grouping effect	—	yes	no	yes

† Fitted values always unique; the coefficient vector need not be when $p > n$ (it is for data in general position).

The one line to remember

On an orthonormal design: OLS b_j; ridge $b_j/(1+\lambda)$; LASSO $\operatorname{sign}(b_j)(|b_j| - t)_+$.
Multiplying never reaches zero; subtracting does.

Objective Formulations & Practical Insights

Sample-Size Scaled Objective

$$\mathcal{L}(\beta_0, \boldsymbol{\beta}) = \frac{1}{2n}\|\mathbf{y} - \beta_0 - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\left[\alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2\right]$$

Matching stationarity conditions with the unscaled objective $\|\mathbf{y}_c - \mathbf{X}_c\boldsymbol{\beta}\|_2^2 + L\|\boldsymbol{\beta}\|_2^2$ yields $L = n\lambda$.

Likewise, the soft-thresholding parameter for LASSO satisfies $t = n\lambda$.

- **Predictor Standardisation:** Compulsory in practice so arbitrary measurement units do not distort penalty severity.
- **Reproducible Data Splits:** Cross-validation folds must be fixed to reliably compare risk profiles.
- **Tuning Parameter α:** Functions as a structural choice between sparsity ($\alpha = 1$) and grouping ($\alpha = 0$), rather than an easily estimated parameter.

Conclusion: which procedure should you use?

OLS

$p \ll n$, predictors not badly collinear, and you need unbiased estimates with classical inference (t/F tests, CIs) — there is nothing to gain from a penalty here.

Ridge

Predictors are correlated or p is close to (or exceeds) n, and you want to keep every predictor in the model — purely prediction-focused, no need for a sparse subset.

LASSO

You believe the true model is sparse and want automatic selection. Caution: among correlated predictors it keeps (almost) one at random and drops the rest.

Elastic net

Default choice when predictors are both numerous and correlated: keeps LASSO's exact zeros, repairs its arbitrary tie-breaking within a correlated group, and stays well-posed whenever $p > n$.

Rule of thumb: start with elastic net, tune (α, λ) by cross-validation; move to ridge if sparsity is wrong for the problem, to LASSO if predictors are near-independent, to OLS only when $p \ll n$ and classical inference matters more than prediction risk.

Open questions

Where the penalty family goes next

- **Best subset** $= \ell_0$ penalty $\Rightarrow$ hard-thresholding $b_j \mathbb{1}(|b_j| > t)$: discontinuous in b_j, combinatorial to compute. LASSO is its convex relaxation.
- **Adaptive LASSO**: weights w_j in $\sum_j w_j |\beta_j|$ to repair LASSO's bias on large coefficients.
- **Group LASSO**: penalise $\|\boldsymbol{\beta}_g\|_2$ per group g to select whole blocks (e.g. all dummies of a factor) together.
- **Degrees of freedom**: $\mathrm{df}(\lambda) = \mathrm{tr}(\mathbf{H}_\lambda)$ for ridge; for LASSO, $\mathbb{E}[\#\{j : \hat{\beta}_j \neq 0\}]$.

Theoretical Exercises

Invariance from **H**; ridge under wild rescaling; $p > n$ and the "at most n" bound; hard vs soft thresholding and continuity; stability of $\hat{\lambda}$ vs stability of the selected set; grouping effect on a block of five correlated predictors.